Curriculum Vitae

Senior Financial Data Engineer · Market & Financial Data Platforms
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> *Financial data that arrives on time, matches its source, and can be explained.*

Profile

Data engineer with 5+ years in industry, mostly on financial products. Designed highly available ClickHouse, MongoDB, and Redis clusters holding **5B+ records** for a credit risk platform with **10M+ users**, and tuned its Spark pipelines. Streamed on-chain transactions through **Kafka** into PostgreSQL and search with sub-second lookups. Moved investment securities SQL from **Oracle to EDB (PostgreSQL)** without changing its business logic. Built the document pipeline for the team that won **FPT AI Hackathon 3 (2025)**.

Looking for Senior Financial Data Engineer roles, with the aim of growing into Staff-level work on platform architecture and shared standards. Started in credit risk modeling, which shapes how I design for the people who use the data. Cares most about point-in-time correctness, reconciliation with the source, careful schema changes, and ingestion that can be safely replayed.

Platform & Architecture Highlights

- **Multi-store architecture:** ClickHouse, MongoDB, and Redis clusters with high availability and disaster recovery for 5B+ records (AZEN Global).
- **Streaming, end to end:** on-chain collection → Kafka → PostgreSQL → OpenSearch / Algolia, searchable in under a second (SupperTree).
- **Database migration:** Oracle → EDB (PostgreSQL), with SQL rewritten to ANSI and the securities logic left unchanged (FPT Software Korea).
- **Throughput:** Spark tuning for a platform with 10M+ users (AZEN Global).
- **Operational visibility:** live monitoring dashboards and reporting APIs that operators and stakeholders used day to day (AZEN Global, FPT Software Korea).

Awards & Recognition

Champion — FPT AI Hackathon 3 · *2025*
First place in the *KT/KB Technical Solutions* track of a global competition across FPT offices, as part of team *Cẩm Y Vệ (FKR-KTX)*. Our entry, **AISOL**, was a Japanese-language enterprise RAG system built on Qdrant, Gemini, and Qwen.

Professional Experience

FPT Software Korea — *Data Software Engineer*

June 2024 – Present · South Korea

Investment Securities Project — Meta

Migration of an investment securities database from Oracle to EDB (PostgreSQL).

- Rewrote Oracle SQL as ANSI SQL so it ran on EDB (PostgreSQL), and made the queries faster in the process.
- Turned complex securities business requirements into efficient SQL.

Technologies: Oracle · EDB (PostgreSQL) · ANSI SQL

QA/QC Data Analytics and Visualization Dashboard

Data processing and reporting for quality operations.

- Built SQL and Java pipelines to aggregate, preprocess, and analyze QA/QC records.
- Used Apache Commons Math for statistical processing and delivered reports through Spring Boot REST APIs and Chart.js dashboards.

Technologies: SQL · Java · Oracle · Apache Commons Math · Chart.js · REST APIs · Spring Boot

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SupperTree — *Software Engineer*

April 2021 – June 2024 · South Korea

BlockStream — Real-Time Blockchain Data System

- Built real-time collection and monitoring of Ethereum and Polygon transactions on a publish–subscribe design.
- Sent transactions through Kafka into PostgreSQL, then indexed them in OpenSearch and Algolia for sub-second search.
- Owned the flow from collection to storage and indexing, including metadata enrichment.

Discord Emotion Tracker

- Built a message ingestion and enrichment pipeline for Discord customer service communications.
- Integrated AWS Comprehend and HuggingFace Rust-BERT for sentiment analysis and AWS Translate for English-to-Korean translation.
- Structured MongoDB storage for messages, translations, and sentiment metrics to support downstream queries.

NFT Management — Token Staking and Earning App

- Built the backend for NFT staking and token rewards, with attention to transaction integrity.
- Connected the app to on-chain NFT transactions and optimized the database queries behind it.

Technologies: Python · Rust · Golang · TypeScript · Node.js · Kafka · AWS Comprehend/Translate · HuggingFace · Algolia · OpenSearch · PostgreSQL · MongoDB · Kinesis

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FitPetMall — *Data Software Engineer*

July 2020 – April 2021 · South Korea

InsightFlow — Customer Behavior Data Pipelines

- Developed data models and workflows for collecting and processing e-commerce customer behavior records.
- Built real-time event streaming and data processing with Kafka and ELK Stack.
- Delivered data workflows for shipping, product delivery, and promotional analysis.

Technologies: Python · FastAPI · ELK Stack · Apache Kafka · Oracle

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AZEN Global — *Data Scientist / ML Engineer*

March 2019 – July 2020 · South Korea

ABACUS — Advanced Credit Risk Prediction Analysis Tool

Data infrastructure and processing for a fault-detection and credit risk platform serving more than **10 million users**.

- Designed highly available ClickHouse, MongoDB, and Redis clusters holding 5B+ records, with disaster recovery in place.
- Reworked fault-detection pipelines on Apache Spark, raising throughput for workflows serving 10M+ users.
- Built live monitoring dashboards and reports so the team could see platform health and act on it.
- Developed machine learning credit risk models that improved scoring accuracy by 3%.

Technologies: Python · Django · FastAPI · ClickHouse · MongoDB · Oracle · Redis · Apache Spark

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National Cancer Center — *Graduate Research Assistant (Health Data Analyst)*

September 2016 – September 2018 · Goyang, South Korea

Mendelian Randomization — Causal Analysis of BMI on Thyroid Cancer Risk

- Integrated genome-wide SNP data and environmental factors for statistical analysis using Python, R, and SAS.
- Applied instrumental variable analysis (Mendelian Randomization) to evaluate causal relationships.

Technologies: SAS · Python · R · Linux

Education

Master of Public Health — Health Data Science
National Cancer Center, Goyang, Republic of Korea · *2016 – 2018*
Thesis: *Causal Effect of BMI on Thyroid Cancer Risk using Mendelian Randomization*

Bachelor of Engineering — Biotechnology
Industrial University of Ho Chi Minh City, Vietnam · *2008 – 2013*

Credentials

- AWS Certified Data Engineer — Associate

- AWS Certified AI Practitioner — Foundational
- Microsoft Certified: Azure AI Fundamentals
- Nanodegree: Data Engineering with Microsoft Azure
- Astronomer Certification: Apache Airflow Fundamentals

Technical Skills

Domain	Tools & Methods
Financial Data Domains	Investment Securities · Credit Risk · On-Chain Transactions · Fault Detection
Streaming & Real-Time Ingestion	Apache Kafka · Kinesis · Publish–Subscribe · ELK Stack
Batch & Distributed Processing	Apache Spark · Airflow · dbt · ETL/ELT · Pandas
Analytical & Operational Storage	ClickHouse · PostgreSQL / EDB · Oracle · MongoDB · Redis
Platform Reliability	Highly Available Clusters · Disaster Recovery · Operational Monitoring Dashboards
SQL Engineering & Migration	Query Optimization · Oracle → EDB (PostgreSQL) Migration · ANSI SQL
Search & Retrieval	OpenSearch · Algolia · Qdrant · Vector Search
AI Data Engineering	Multimodal Document Processing · RAG · LangChain · Gemini · Qwen · HuggingFace
Delivery	FastAPI · Django · Spring Boot · REST APIs
Cloud & Infrastructure	AWS · Azure · Docker · Kubernetes
Languages	Python · SQL · Golang · Rust · Java · TypeScript

Open Source & Side Projects

- **discord-playdapp-bot** (https://github.com/augustine0890/discord-playdapp-bot) — Rust · Serenity · MongoDB · Docker
  High-performance Discord bot built in Rust using Serenity for gaming community management. Handles tournament ticket exchange, player engagement tracking, and leaderboard ranking via slash commands.
- **outfit-square** (https://github.com/augustine0890/outfit-square) — Python · Discord.py · MongoDB · Poetry
  Python-based Discord bot for user point management in metaverse gaming channels. Integrates MongoDB for persistent storage with multi-stage (dev/prod) deployment and OAuth2 authentication.

Personal Details

- Citizenship: Vietnamese
- Country of Residence: Republic of Korea
- Languages: English (fluent) · Vietnamese (native)